



UT Dallas 2026 Undergraduate Catalog

School of Natural Sciences and Mathematics

Certificate in Biomedical Sciences

The post-baccalaureate Certificate in Biomedical Sciences (CBioMed) is offered through the School of Natural Sciences and Mathematics (NSM) and administered through the Health Professions Advising Center (HPAC). A rigorous curriculum allows students to further develop their scientific knowledge in preparation for application to schools of medicine, dentistry, or podiatry. Program requirements also include clinical, community service and/or research hours, independent from course credit and initiated by the student. Certificate students access HPAC services receiving assistance with the application process.

Application for the program is through the [ApplyTexas](https://www.utdallas.edu/admissions) online application at www.utdallas.edu/admissions. Applicants apply as "Transfer, Undergraduate" students in the School of Natural Sciences and Mathematics, and select the "Undergraduate Certificate in Biomedical Sciences." A supplemental application, as well as the booklet "Information and Program Guidelines," can be found on the [HPAC web-page](#). Please contact the HPAC office for deadlines in submitting the supplemental application.

Admission Requirements

Prospective students interested in enrolling in the Certificate in Biomedical Sciences program will be considered for admission based on the following standards:

- met University admission requirements established for transfer undergraduate students;
- earned a bachelor's degree from a U.S. college or university;
- exhibited clear motivation for a career in medicine, dentistry, or podiatry (as evidenced by previous coursework, clinical exposure and/or a realistic plan for preparation);
- completed the CBioMed program supplemental application; and,
- earned an undergraduate grade point average (GPA) of at least 2.750.

Note: Competitive applicants for the CBioMed program should have completed, or be in the process of completing, an introductory sequence - for science majors - of chemistry, biology and physics.

Program Requirements

The certificate program is designed for students who are preparing for entrance into a medical, dental or podiatry school.

Requirements for completion of the Certificate in Biomedical Sciences program include:

- A minimum of 24 post-baccalaureate undergraduate semester credit hours of approved courses at UT Dallas.
- Of the 24 semester credit hours completed toward the certificate, a minimum of 9 semester credit hours must be HPAC advisor approved upper-division science courses.
- In addition to the science courses, students must complete at least one course with content covering health disparities, professionalism, and/or ethics.
- Completion of all admission prerequisite courses for the health profession schools to which the student will be applying.
- A UT Dallas post-baccalaureate GPA of at least 3.300.
- Evidence of at least 50 clock hours of approved clinical, community service and/or research activities documented according to program standards.
- Completion of the Health Professions Evaluation (HPE) Process and recommendation by the HPAC Advisory Committee.

Curriculum

A variety of classes are available to students, depending on their particular needs and previous experience in undergraduate science courses. Students are required to work with an HPAC advisor in order to plan their curriculum for the program. HPAC advisors work with students to develop a curricular plan that is based on their individual circumstances, including past academic history and career goals. Courses that may be included to fulfill the certificate program requirements are listed below. Not all courses are taught every semester.

Biology

- BIOL 2311** Introduction to Modern Biology I
- BIOL 2111** Introduction to Modern Biology Workshop I
- BIOL 2312** Introduction to Modern Biology II
- BIOL 2112** Introduction to Modern Biology Workshop II
- BIOL 3401** Genetics
- BIOL 3402** Molecular and Cell Biology
- BIOL 3303** Introduction to Microbiology
- BIOL 3305** Evolutionary Analysis
- BIOL 3318** Forensic Biology
- BIOL 3320** Applied Genetics
- BIOL 3335** Microbial Physiology
- BIOL 3336** Protein and Nucleic Acid Structure
- BIOL 3355** Clinical Pathophysiology
- BIOL 3357** Mammalian Physiology with Lab

BIOL 3461 Biochemistry I

BIOL 3462 Biochemistry II

BIOL 3370 Exercise Physiology

BIOL 3380 Biochemistry Laboratory

BIOL 3385 Medical Histology

BIOL 3455 Human Anatomy and Physiology with Lab I

BIOL 3456 Human Anatomy and Physiology with Lab II

BIOL 4310 Cellular Microbiology

BIOL 4341 Genomics

BIOL 4345 Immunobiology

BIOL 4350 Medical Microbiology

BIOL 4353 Molecular Biology of HIV/AIDS

BIOL 4357 Molecular Neuropathology II

BIOL 4366 Molecular Biology of Cancer

BIOL 4385 Oral Histology and Embryology

BIOL 4V40 Special Topics in Molecular and Cell Biology [when topic is Oral Histology]

Chemistry

CHEM 1311 General Chemistry I

CHEM 1111 General Chemistry Laboratory I

CHEM 1312 General Chemistry II

CHEM 1112 General Chemistry Laboratory II

CHEM 2123 Introductory Organic Chemistry Laboratory I

CHEM 2125 Introductory Organic Chemistry Laboratory II

CHEM 2323 Introductory Organic Chemistry I

CHEM 2325 Introductory Organic Chemistry II

CHEM 2401 Introductory Quantitative Methods in Chemistry

CHEM 3321 Physical Chemistry I

CHEM 3322 Physical Chemistry II

CHEM 4381 Green Chemistry and Green Fuels

Neuroscience

NSC 3361 Introduction to Neuroscience

NSC 4351 Medical Neuroscience

NSC 4352 Cellular Neuroscience

NSC 4354 Integrative Neuroscience

NSC 4356 Neurophysiology

NSC 4358 Neuroscience of Pain

NSC 4362 Molecular Neuroscience

NSC 4363 Neuropharmacology

NSC 4366 Neuroanatomy

NSC 4367 Developmental Neurobiology

NSC 4371 Neural Plasticity

NSC 4373 Sensory Neuroscience

Physics

PHYS 3330 Numerical Methods in Physics and Computational Techniques

Statistics

STAT 2332 Introductory Statistics for Life Sciences

Other Disciplines

GEOG 3357 Spatial Dimensions of Health and Disease

GEOS 2324 Energy, the Environment and Human Health

All certificate students are required to take, as a part of their program curriculum, a class covering topics in health disparities, professionalism and/or ethics.

Elective Courses

ECON 3330 Economics of Health

GEOG 3357 Spatial Dimensions of Health and Disease

HIST 3328 History and Philosophy of Science and Medicine

HLTH 1100 Career Explorations for the Health Professions

HLTH 1322 Human Nutrition

HLTH 3101 Medical Terminology

HLTH 3300 Pre-Health Professional Development

HLTH 3305 The U.S. Healthcare System

HLTH 4380 Special Topics in Healthcare

HMGH 3301 Introduction to Healthcare Management

PHIL 3320 Medical Ethics

PHIL 4321 Philosophy of Medicine

PSY 2301 Introduction to Psychology

PSY 4346 Human Sexuality

PSY 4328 Health Psychology

SOC 1301 Introduction to Sociology

SOC 4369 Public Health and Society

SOC 4371 Mental Health and Illness

SOC 4372 Health and Illness

SPAN 3330 Medical Spanish

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